

# 1.6 Ethical, legal, cultural and environmental impacts of digital technology

Revise360

Teacher guide  
Unit 1.6  
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Where this unit sits, what it covers, and how far

## Where this unit sits

|                                    |                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification                      | GCSE Computer Science (9-1), specification code J277, from 2020                                                                                                                                                                              |
| Component                          | Component 01: Computer systems — 80 marks, 1 hour 30 minutes                                                                                                                                                                                 |
| What OCR says the component covers | the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software                                                                    |
| This unit                          | topic 1.6 Ethical, legal, cultural and environmental impacts of digital technology                                                                                                                                                           |
| Checked                            | 3 October 2026, at <a href="https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/">https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/</a> |

**How far this has been checked.** The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

**Before it:** 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software

**After it:** 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

## The lesson sequence and the questions it answers

| # | LESSON                            | THE BIG QUESTION                                                             | ACTIVITIES | MARKS |
|---|-----------------------------------|------------------------------------------------------------------------------|------------|-------|
| 1 | <b>Impact desk</b><br>e1-101      | What should be considered before a new technology is introduced?             | 13         | 33    |
| 2 | <b>Privacy lab</b><br>e1-102      | What privacy issues does computing create, and where should the limits be?   | 13         | 25    |
| 3 | <b>Law court</b><br>e1-103        | What does computing legislation prohibit?                                    | 12         | 25    |
| 4 | <b>Culture street</b><br>e1-104   | How does computing affect how people live and work?                          | 13         | 30    |
| 5 | <b>Planet works</b><br>e1-105     | What is the environmental impact of computing, across a device's whole life? | 13         | 29    |
| 6 | <b>Stakeholder room</b><br>e1-106 | How does digital technology affect society locally, nationally and globally? | 14         | 27    |

| # | LESSON                                  | THE BIG QUESTION                                                           | ACTIVITIES | MARKS |
|---|-----------------------------------------|----------------------------------------------------------------------------|------------|-------|
| 7 | <b>Licence lab</b><br>e1-l07            | What are the implications of choosing open source or proprietary software? | 13         | 28    |
| 8 | <b>End of topic test</b><br>e1-l08-test | assesses the whole unit                                                    | 0          | 0     |
| 9 | <b>Name that Act</b><br>e1-bonus        | stretch work, no new content                                               | 1          | 0     |

## Prior knowledge, vocabulary and skills

**Assumed before the unit:** the outcomes of 1.1, 1.2, 1.3, 1.4, 1.5

**Vocabulary introduced:** Ethical issue, Stakeholder, Personal data, Consent, Data Protection Act 2018, Computer Misuse Act 1990, Digital divide, Globalisation, E-waste, Carbon footprint, Trade-off, Open source, Proprietary

**What a pupil does, not only reads:** impact, lawgame, match, mcq, multi, order, sort

## Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

**How this was worked out.** The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage . json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

| OUTCOME                                                                               | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|---------------------------------------------------------------------------------------|--------|--------|-----------|----------|----------|
| know the range of issues to consider when a new technology is introduced<br>e1-l01-o1 | e1-l01 | yes    | yes       | yes      | Full     |
| understand at least one ethical issue of computer technology<br>e1-l01-o2             | e1-l01 | yes    | yes       | yes      | Full     |
| be able to argue both sides and reach a conclusion<br>e1-l01-o3                       | e1-l01 | yes    | yes       | —        | Partial  |
| understand at least one issue related to privacy and computer technology<br>e1-l02-o1 | e1-l02 | yes    | yes       | yes      | Full     |
| be able to weigh the benefits and risks of collecting personal data<br>e1-l02-o2      | e1-l02 | yes    | yes       | yes      | Full     |
| know the principles of the Data Protection Act 2018<br>e1-l03-o1                      | e1-l03 | yes    | yes       | yes      | Full     |
| know the principles of the Computer Misuse Act 1990<br>e1-l03-o2                      | e1-l03 | yes    | yes       | yes      | Full     |
| know the principles of the Copyright, Designs and Patents Act 1988<br>e1-l03-o3       | e1-l03 | yes    | yes       | yes      | Full     |
| understand the impact of technology on daily life<br>e1-l04-o1                        | e1-l04 | yes    | yes       | yes      | Full     |
| understand the digital divide<br>e1-l04-o2                                            | e1-l04 | yes    | yes       | yes      | Full     |

| OUTCOME                                                                                | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|----------------------------------------------------------------------------------------|--------|--------|-----------|----------|----------|
| understand globalisation<br>e1-l04-o3                                                  | e1-l04 | yes    | yes       | yes      | Full     |
| understand the environmental impact of manufacturing computers<br>e1-l05-o1            | e1-l05 | yes    | yes       | yes      | Full     |
| understand the environmental impact of using computers<br>e1-l05-o2                    | e1-l05 | yes    | yes       | yes      | Full     |
| understand the environmental impact of disposing of computers<br>e1-l05-o3             | e1-l05 | yes    | yes       | yes      | Full     |
| be able to identify key stakeholders<br>e1-l06-o1                                      | e1-l06 | yes    | yes       | yes      | Full     |
| be able to consider a scenario from the perspective of stakeholders<br>e1-l06-o2       | e1-l06 | yes    | yes       | yes      | Full     |
| understand impacts at local, national and global levels<br>e1-l06-o3                   | e1-l06 | yes    | yes       | yes      | Full     |
| understand the difference between open source and proprietary software<br>e1-l07-o1    | e1-l07 | yes    | yes       | yes      | Full     |
| understand the implications of using open source and proprietary software<br>e1-l07-o2 | e1-l07 | yes    | yes       | yes      | Full     |

**18 full** · **1 partial** · **0 not covered**, of 19 outcomes.

## Gaps, assumptions and overlap

- **e1-l01-o3** be able to argue both sides and reach a conclusion — nothing in this unit assesses it

### Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- e1-l03-x3 A company finds that an employee has copied its customer database and taken it to a competitor. Explain which laws may have been broken.
- e1-l05-x2 Explain why electronic waste is a problem when sent to landfill.

### What could not be traced

The wall text of e1-l01, e1-l02, e1-l03, e1-l04, e1-l05, e1-l06, e1-l07, e1-bonus is painted into the 360 image and held nowhere as text, so an outcome in those lessons lesson can only be traced through its activities and its exam questions. Read the rows above for them with that in mind.

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